

not too high. Above all, these are predominantly battery-operated portable devices. So these need low-power communication technology for conserving battery.

The second class of devices is the ones that support a full IPv6 protocol and can communicate to central application servers. These devices can also be Gateways, wherein these sport two different types of communication ports. One of these work with Things, which use non-IPv6 data communication and the second support data communications ports like Wi-Fi, DSL and FTTH on the other side towards application servers. These use more powerful processors and are not constrained by battery power. In addition to gateway functions, these also support multiple sensors.

Based on these functionalities, IoT devices are classified into constrained devices and unconstrained devices. Constrained devices have restrictions on power supply and memories (RAM and flash). These are further classified into three classes (refer table).

Unique elements of the IoT

There are a couple of things that are very unique to the IoT. First is IPv6 protocol. It is an upgrade to IPv4, which ran out of options to allocate unique addresses and hence IPv6 was proposed and accepted. With 7-byte- or 128-bit-long addresses, there are enough unique addresses available to allocate to the IoT.

Second unique element is the concept of Delay Tolerant Networks (DTN). It is designed for intermittent connectivity where there is no direct path from the source to destination. DTN was originally developed for NASA for interplanetary communication, which is intermittent and is a common feature in space communication. With TCP/IP being a connection-oriented protocol, NASA was looking for an alternative and DTN was developed for the same.

DTN helps IoT applications, too. As Things with simpler sensors are not connected to the central server all the time, these collect and store data till it gets connected to another node through which the data passes. The recipient node collects data and forwards it through the network till it reaches the central server.

Compared to regular networks, IoT networks can have large variable delays, which need to be addressed, and DTN solves the problem.

Embedded systems for the IoT

The IoT is an evolving technology with standards getting evolved simultaneously, so it is difficult to define a fixed architecture for the IoT. Second, unlike standard embedded systems, the IoT is a solution (in most cases) and the device alone cannot solve the problem. This is primarily due to the large volume of data being collected from sensors, and making sense of the data needs application software. In this scenario, embedded system designers need to understand the following points when they design a system:

Understanding user needs and the solution output. For example, if a Thing has to measure the temperature of the body for a fitness application, the designer needs to know, apart from the temperature sensor, amount of data that needs to be measured and stored, and the amount of battery power that is needed. These will facilitate the processor selection as well as the resources needed in the processor.

Communicating with the central server. Whether the server will communicate through the Gateway or directly to the server will decide the CPU power needed as well as memory resources needed to implement the solution.

Interconnection. This offers the path between Things and the central application. Today, most telecom service providers have designed their services to offer C2B

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